

DIFFERENTIATION: SPEED CHECKER

Class 12 Applied Maths — MCQ Shortcut Magic — By Thunderstudy

Function = Position | First Derivative = Speed | Second Derivative = Acceleration.
Don't solve—spot the pattern!

SUPER FAST MCQ SHORTCUT TRICKS

✓ 1. Power Rule = Instant Answer

Shortcut:

$$d/dx(x^n) = n \cdot x^{n-1}$$

Hack: Reduce power by 1 and multiply by the old power. **Time saved: 20 sec**

✓ 2. Constant Killer Rule ✨

Any constant alone disappears:

$$d/dx(7) = 0$$

MCQ Shortcut: If an option has a constant derivative $\neq 0 \rightarrow$ **✗ Eliminate immediately.**

✓ 3. Quotient Rule Elimination Trick

Fractions:

$$(vu' - uv') / v^2$$

FAST CHECK: The denominator is **ALWAYS** squared (v^2). If an option doesn't have it → eliminate instantly.

✓ 4. Chain Rule Detector

Function inside function? Two steps: **1. Differentiate outside. 2. Multiply by derivative of inside.**

Example: $d/dx(3x+1)^5 = 5(3x+1)^4 \times 3$. The inside derivative **MUST** appear!

✓ 5. Maxima & Minima SUPER FAST ★

Step 1: Set $f'(x) = 0$. Step 2: Check $f''(x)$:

- Positive (+) = **Minimum**
- Negative (−) = **Maximum**

💥 90% of MCQs use this rule only.

✓ 6. Tangent & Normal Instant Trick

Slope of tangent (m) = dy/dx . At a point? Just substitute the value. No long equation needed first.

✓ 7. MOST POWERFUL MCQ ELIMINATION TRICK 🔥

Before solving, check these four things to remove 2-3 options without calculation:

- ✓ Check Degree (Power)
- ✓ Check Sign (+/-)
- ✓ Check Denominator Square
- ✓ Check Chain Multiplication

⚡ 5-SECOND MCQ STRATEGY (Boards Topper Method)

Match pattern ≠ Solve fully.

1. Look for rule → 2. Apply pattern → 3. Match option (don't simplify!)

Example (Boards Level)

Find derivative of $(2x+3)^4$.

Outside: $4(2x+3)^3$ | Inside: $\times 2$

✓ **Answer: $8(2x+3)^3$**

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